

Teamwork and Individual Productivity

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Introduction & Motivation

- **Motivation:** Scientific research, professional sports, and modern business are increasingly collaborative (Wuchty et al. 2007).
- **Question:** How do we evaluate individual productivity when we only observe collective team success?
- **The Paradox:**
 - 1 To measure *individual productivity*, we must know how much an individual contributes to her teams.
 - 2 To determine *individual contribution*, we must know her productivity (more productive members bring more skill, visibility, and value).
- **Solution:** Join these problems into a single, endogenous measure called **CoScore**
- **Insight:** Your credit on a project should depend on the strength of your teammates. Stronger teammates mean you receive less credit, while weaker teammates mean you receive more.

Definition of the CoScore

Let N be the set of individuals, and C be the database of projects.

- Each project $p \in C$ has a team $T(p) \subseteq N$ and a positive worth $w(p) > 0$.
- Each individual $i \in N$ has a production record $C_i \subseteq C$ of projects she participated in.

Definition: The CoScore System

The productivity score s_i of individual $i \in N$ is defined by:

$$s_i = \sum_{p \in C_i} w(p) \frac{s_i}{\sum_{j \in T(p)} s_j} \quad \forall i \in N \quad (1)$$

CoScore is well-defined if each individual has at least one solo project with $w(p) > 0$. Existence follows from Brouwer's fixed-point theorem; uniqueness follows from the strict concavity of an underlying potential function.

Comparison of the approaches

Egalitarian score

Credit is allocated equally among all members:

$$s_i(p) = \frac{1}{\#T(p)} w(p)$$

- Teammates are treated as anonymous entities; their records do not affect your credit.
- Individual scores can be calculated in isolation, ignoring the global network.
- Fails to recognize when a team member carries a team or when a weak teammate is carried.

CoScore

Credit is allocated proportionally to productivity:

$$s_i(p) = \frac{s_i}{\sum_{j \in T(p)} s_j} w(p)$$

- Your credit depends on your teammates' productivity, which depends on their other teammates' productivity, which depend on...
- Uses the entire database of collaborations and success rates simultaneously.

Axioms

To understand credit allocation rules formally, let a rule r recommend a credit allocation $r(D)$ for any database D such that $\sum_{i \in T(p)} r_i(p, D) = w(p, D)$ for each project p .

- **Axiom 1: Anonymity**

The names or tags of individuals do not matter. Permuting the production records of individuals permutes their allocated credits accordingly.

- **Axiom 2: Own Record Only**

An individual's credit on a project depends only on her own production record, not on the records of individuals who do not collaborate with her.

Theorem 1:

A credit allocation rule satisfies **Anonymity** and **Own Record Only** if and only if it is the **egalitarian rule**.

The Axioms

To allow an individual's credit to depend on her teammates' strength, we must drop *Own Record Only*. CoScore is characterized by three alternative axioms:

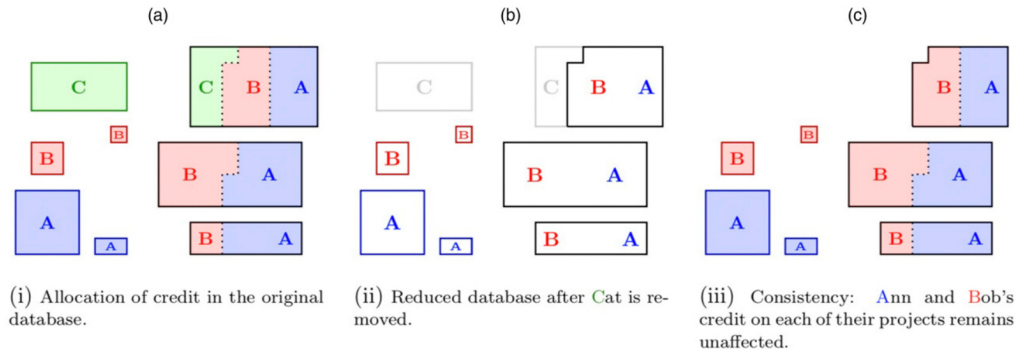
- **Axiom 3: Consistency** — Removing an individual k and reducing collaborative projects by her credit $r_k(p)$ does not affect the remaining credit allocations.
- **Axiom 4: Aggregate Worth** — Merging projects of the same team into a single project of equal aggregate worth does not affect credit.
- **Axiom 5: Aggregate Strength** — Replacing a group of teammates sharing identical collaborative projects with a single “aggregate” teammate does not affect outsider credits.

Theorem 2:

A credit allocation rule satisfies **Consistency**, **Aggregate Worth**, and **Aggregate Strength** if and only if it is the **CoScore rule**.

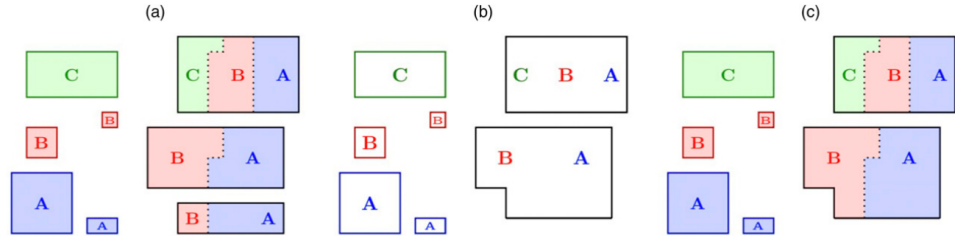
Consistency Axiom

Figure 3. (Color online) Consistency



Aggregate Worth Axiom

Figure 4. (Color online) Aggregate Worth



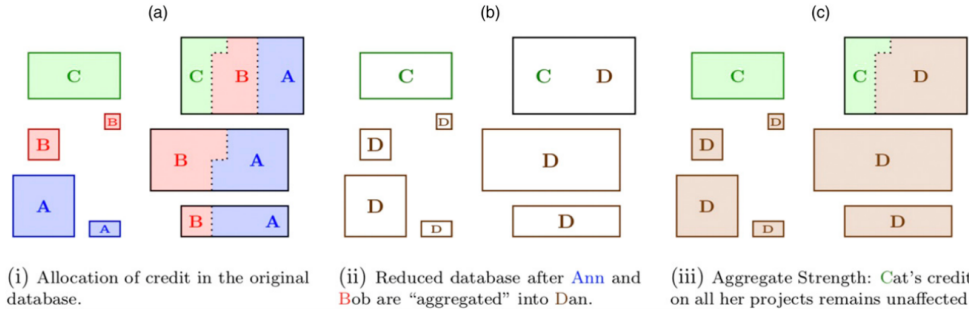
(i) Allocation of credit in the original database.

(ii) Reduced database after Ann and Bob's two joint projects are aggregated into a single project of equal total worth.

(iii) Aggregate Worth: Ann and Bob's credit on the aggregate project are equal to the total credit they were allocated on each of their two joint projects. The allocation of credit on all other projects remains unaffected.

Aggregate Strength Axiom

Figure 5. (Color online) Aggregate Strength



Egalitarian vs. CoScore

The difference between these two paradigms boils down to a choice of axioms:

Axiom	Egalitarian Rule	CoScore Rule
Anonymity	Satisfied	Satisfied
Own Record Only	Satisfied (Evaluated in Isolation)	Violated
Consistency	Satisfied	Satisfied
Aggregate Worth	Satisfied	Satisfied
Aggregate Strength	Violated (Ignores Teammate Quality)	Satisfied (Adjusts for Strength)

Table: Axiomatic Comparison of Credit Allocation Rules

- The egalitarian rule guarantees that your score is immune to what others do outside your projects.
- CoScore sacrifices isolation to ensure that your credit correctly reflects the relative quality of your co-workers.

The Generalized α -CoScore

In real-world applications we encounter two challenges:

- 1 **Varying Career Lengths:** Senior individuals naturally accumulate more projects.
- 2 **Varying Participation Levels:** Teammates do not always contribute equal effort/time.

Let A_i represent the career length (age) of individual i , and $t_i(p) > 0$ represent her time/effort participation in project p .

Definition: The Generalized α -CoScore

For a parameter $\alpha \in [0, 1]$, the career-average productivity score s_i satisfies:

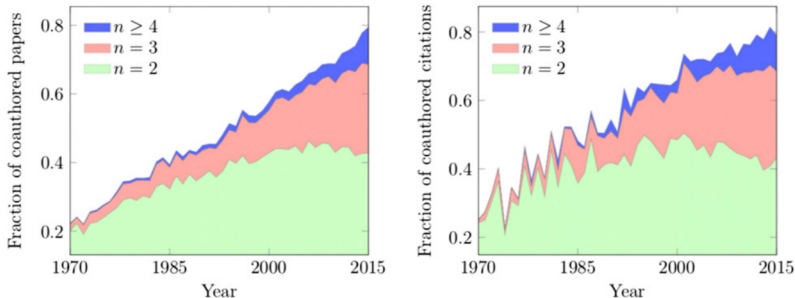
$$s_i = \frac{1}{A_i} \sum_{p \in C_i} w(p) \left[\alpha \frac{t_i(p)}{\sum_{j \in T(p)} t_j(p)} + (1 - \alpha) \frac{t_i(p)s_i}{\sum_{j \in T(p)} t_j(p)s_j} \right] \quad \forall i \in N \quad (2)$$

Economic Interpretation of α :

- $\alpha = 1$: Egalitarian case (credit is based purely on observed time participation).
- $\alpha = 0$: Endogenous case (credit is based purely on endogenous quality/contribution).
- *Low* α fits task where individual effort matters more; *high* α fits team-synergy or weakest-link tasks.

Application 1: Academic Researchers

Figure 6. (Color online) Fraction of Coauthored Papers (Left) and Citations (Right) with Two, Three, and Four or More Authors from 1970 to 2015



Application 1: Academic Researchers

The Coauthorship Network

- **Context:** In economics, coauthors are overwhelmingly listed alphabetically, leaving individual contributions unobserved.
- **Data:** Web of Science database from 33 top economics journals (1970–2015). 73,732 articles and 29,226 unique authors.
- **Definitions:** Project is a paper; worth $w(p)$ is the citation count; participation is binary; A_i is career length.
- **The Trend:** Coauthored papers rose from **20%** in 1970 to **80%** in 2015.

Computation

The generalized α -CoScore is computed by iterating:

$$\varphi_i(x) = \frac{1}{A_i} \sum_{p \in C_i} w(p) \left[\frac{\alpha}{\#T(p)} + \frac{(1-\alpha)x_i}{\sum_{j \in T(p)} x_j} \right]$$

- Shows a divide between $\alpha < 1$ and the egalitarian benchmark ($\alpha = 1$).
- Average rank shift is **2,345 positions** for $\alpha = 0.5$, and average score change is **37.3%**.

Determinants of CoScore in Economics

To see what drives CoScore, the authors run OLS regressions of the allocated share of credit on key metrics:

Independent Variable	0.1-CoScore	0.5-CoScore	0.9-CoScore
# Coauthors/Paper	-0.079***	-0.094***	-0.040***
# Coauthors	0.005***	0.002***	0.0003***
$\log(Egal_i / EgalN_i)$	0.150***	0.077***	0.016***
Solo Citation Share	0.153***	0.060***	0.013***
Observations	12,134	12,134	12,134
Adjusted R^2	0.724	0.818	0.760

Table: OLS Regressions: Share of Credit under Generalized α -CoScore

Key Takeaways: Lower values of α (closer to pure CoScore) strongly favor authors who:

- 1 Have high individual quality as evidenced by solo papers (*Solo Citation Share*).
- 2 Collaborate with multiple different groups (*# Coauthors*).
- 3 Work with relatively weaker coauthors ($\log(Egal_i / EgalN_i)$).

Economics Case Studies: Two Examples

Author A:

- **Record:** 74 papers, 8,113 citations, 66 coauthors. Only **0.1%** of citations are from solo papers.
- **Egalitarian Rank:** 89th
- **0.1-CoScore Rank:** 41st
- **Why?** Author A collaborated successfully with a very large and diverse network of coauthors. CoScore rewards this as proof of independent capability.

Author B:

- **Record:** 73 papers, 7,035 citations, 27 coauthors. Over **50%** of citations are from solo papers.
- **Egalitarian Rank:** 59th
- **0.1-CoScore Rank:** 97th
- **Why?** Several of Author B's highly cited joint papers were coauthored with **Author C** (a Nobel laureate ranked 3rd/4th globally). CoScore endogenously attributes the credit of these joint papers mostly to the Author C.

Application 2: NBA Players

How does CoScore apply to team sports where solo work is impossible?

- **Context:** NBA provides a closed pool of players and teams. Wins are the observable output, but individual contributions are unobserved.
- **Data:** Database of all NBA team-seasons from **1951 to 2011**. 1,352 team-seasons (projects) and 3,800 different players.
- **Definitions:**
 - ▶ Project worth $w(p)$ = number of team wins in that season.
 - ▶ Participation $t_i(p)$ = total minutes played on the court by player i in that season.
 - ▶ Age A_i = player's career span in years.
- **Results:** Only a small fraction of players (11.5% to 14.2%) increase their score compared to the egalitarian baseline. This confirms the existence of "MVPs" who win games regardless of teammate quality.

How CoScore Differs from other measures

CoScore could be considered similar to network centrality measures. It is however not the case.

- **Network Centrality (e.g., Eigenvector Centrality, PageRank):**

- ▶ Your centrality increases if you are connected to other central nodes (“it is good to have successful neighbors”).
- ▶ Defined on pairwise, binary networks of edges.

- **CoScore:**

- ▶ Your credit on a team project **decreases** when your teammates perform better
- ▶ Project worth is finite; credit must be partitioned among members.

Broader Applications

The underlying logic of CoScore can be applied to any context where collective success can be decomposed:

- **Product Bundling and Content Pricing:**

- ▶ Entertainment platforms (Spotify, Netflix, iTunes) bundle multiple songs or movies.
- ▶ By observing subscription revenues of various bundles, CoScore can endogenously determine the individual value contribution of each piece of content.
- ▶ Similarly, museum passes or public transit cards can use CoScore to fairly compensate individual transit operators or museums.

- **Sports Broadcasting Revenues:**

- ▶ Sharing TV broadcasting revenues among teams can be an issue.
- ▶ TV revenue of a game is the joint output of two teams. CoScore can determine each team's underlying drawing power (worth) from the pairwise game records.

Conclusion

- **New framework for measuring teamwork**

CoScore resolves the paradox of teamwork by co-scoring individual productivity and credit simultaneously in a unified fixed-point system.

- **The Axioms:**

CoScore is uniquely characterized by: Consistency, Aggregate Worth, and Aggregate Strength axioms.

- **Decoupling Collaboration Bias:**

By looking at the entire network history, CoScore rewards independent performers while identifying individuals carried by better team members.

- **Extensibility**

Extendable adapt to careers of varying lengths, varying effort/time participations and diverse contexts.